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Fusion model integrating multi-sequence MRI radiomics and habitat imaging for predicting pathological complete response in breast cancer treated with neoadjuvant therapy

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Abstract

Background This study aimed to develop a predictive model integrating multi-sequence MRI radiomics, deep learning features, and habitat imaging to forecast pathological complete response (pCR) in breast cancer patients undergoing neoadjuvant therapy (NAT).

Methods A retrospective analysis included 203 breast cancer patients treated with NAT from May 2018 to January 2023. Patients were divided into training ($n = 162$) and test ($n = 41$) sets. Radiomics features were extracted from intratumoral and peritumoral regions in multi-sequence MRI (T2WI, DWI, and DCE-MRI) datasets. Habitat imaging was employed to analyze tumor subregions, characterizing heterogeneity within the tumor. We constructed and validated machine learning models, including a fusion model integrating all features, using Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves, decision curve analysis (DCA), and confusion matrices. Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) analyses were performed for model interpretability.

Results The fusion model achieved superior predictive performance compared to single-region models, with AUCs of 0.913 (95% CI: 0.770–1.000) in the test set. PR curve analysis showed improved precision-recall balance, while DCA indicated higher clinical benefit. Confusion matrix analysis confirmed the model's classification accuracy. SHAP revealed DCE_LLL_DependenceUniformity as the most critical feature for predicting pCR and PC72 for non-pCR. LIME provided patient-specific insights into feature contributions.

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Conclusion Integrating multi-dimensional MRI features with habitat imaging enhances pCR prediction in breast cancer. The fusion model offers a robust, non-invasive tool for guiding individualized treatment strategies while providing transparent interpretability through SHAP and LIME analyses.

Keywords Breast cancer, Neoadjuvant therapy, Magnetic resonance, Radiomics, Heterogeneity

Introduction

Breast cancer (BC) management now includes surgery, radiotherapy, chemotherapy, endocrine therapy, and targeted therapy [1]. Neoadjuvant therapy (NAT) has become the standard of care for locally advanced BC, aiming to downstage tumors, expand surgical options, and reduce the risks of recurrence and metastasis to improve long-term outcomes. Consequently, primary surgery is no longer considered the optimal treatment option [2–4]. However, BC exhibits high intratumoral heterogeneity (ITH), resulting in vastly different responses to NAT. Although most patients can benefit from NAT, a small subset exhibits a poor response to NAT and must be treated with modified chemotherapy regimens in subsequent treatment. This variability presents a significant clinical challenge and may compromise the long-term survival rate [5, 6]. The effective stratification of patients for optimal treatment decisions remains elusive yet essential in clinical practice. Early assessment and prediction of NAT response are essential for guiding subsequent treatment strategies. The gold standard for assessing pathological complete response (pCR) is through pathological examination of post-NAT surgical specimens, which is inherently limited by time delays and the inability for dynamic monitoring. Non-invasive, real-time, and accurate preoperative assessment of pCR remains a challenge. The current in-practice evaluation of NAT response is routinely guided by the Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 [7]. However, due to BC's complex heterogeneity, reliance solely on the tumor's maximum diameter as a measure of treatment effect can be misleading and subject to variability among observers [8]. This approach may also overlook critical microscopic information within the tumor.

Breast MRI is recommended for detecting tumor response to NAT. Medical imaging-based machine learning models show great potential in accurately predicting pCR before surgery. Radiomics features extracted from multiparametric MRI can detect tumor response to NAT in BC [9–11]. Previous studies have predominantly focused on intratumoral features to assess the NAT response in BC [12–15]. However, these approaches often neglect the tumor's complex internal architecture and surrounding microenvironment, including changes in vascular density induced by NAT. These changes lead to variations in heterogeneity among different patients during the treatment of NAT. In this context, a simplistic reliance on intratumoral features for characterizing

the heterogeneous nature of BC is inadequate. Therefore, increasing effort has been dedicated to characterizing the tumor heterogeneity with the peritumoral region and habitat imaging as methods [16–18]. Unlike traditional radiomics approaches that treat the tumor as a monolithic entity, habitat imaging divides the tumor into subregions, often referred to as 'habitats'. These subregions, characterized by parameter variations such as blood flow, cell density, and necrosis, have distinct prognostic implications [17, 19]. Furthermore, habitat imaging provides novel insights into how tumor heterogeneity manifests at the imaging level and has demonstrated potential for enhancing the assessment of NAT efficacy [20].

This retrospective study aims to develop a radiomics model based on multiparametric MRI images, incorporating tumor heterogeneity parameters, to improve pCR prediction in BC patients undergoing NAT. We hypothesize that the fusion of these imaging biomarkers will enhance predictive accuracy over conventional radiomics approaches. Additionally, the study will conduct an interpretability analysis of the integrated model's predictive logic. The workflow of the study is briefly outlined in Fig. 1, summarizing the key steps involved. The figure was created by Figdraw.

Materials and methods

Study population

Patients with newly diagnosed BC at our institution from May 2018 to January 2023 were initially included in this study. The study was approved by the Huazhong University of Science and Technology Tongji Medical College Affiliated Tongji Hospital Medical Ethics Committee (Approval No. TJ-IRB20231173). The inclusion criteria were as follows: (1) Patients with primary lesion(s) confirmed as breast cancer via core needle biopsy prior to NAT; (2) Patients who underwent standard NAT before surgery and received a breast MRI approximately one week prior to the initiation of NAT; (3) Patients with complete clinical, pathological, and imaging data. The exclusion criteria were: (1) Patients with a history of other malignant tumors (s); (2) Patients with incomplete or non-standardized NAT treatment or surgery; (3) Patients with poor image quality, indistinct lesion presentation, or difficulty in ROI delineation. Finally, a total of 203 patients were enrolled in the study for further analysis. Subsequently, patients were allocated to the training and test sets in a ratio of 8:2 using computer-generated randomization. Specifically, 162 patients were assigned to

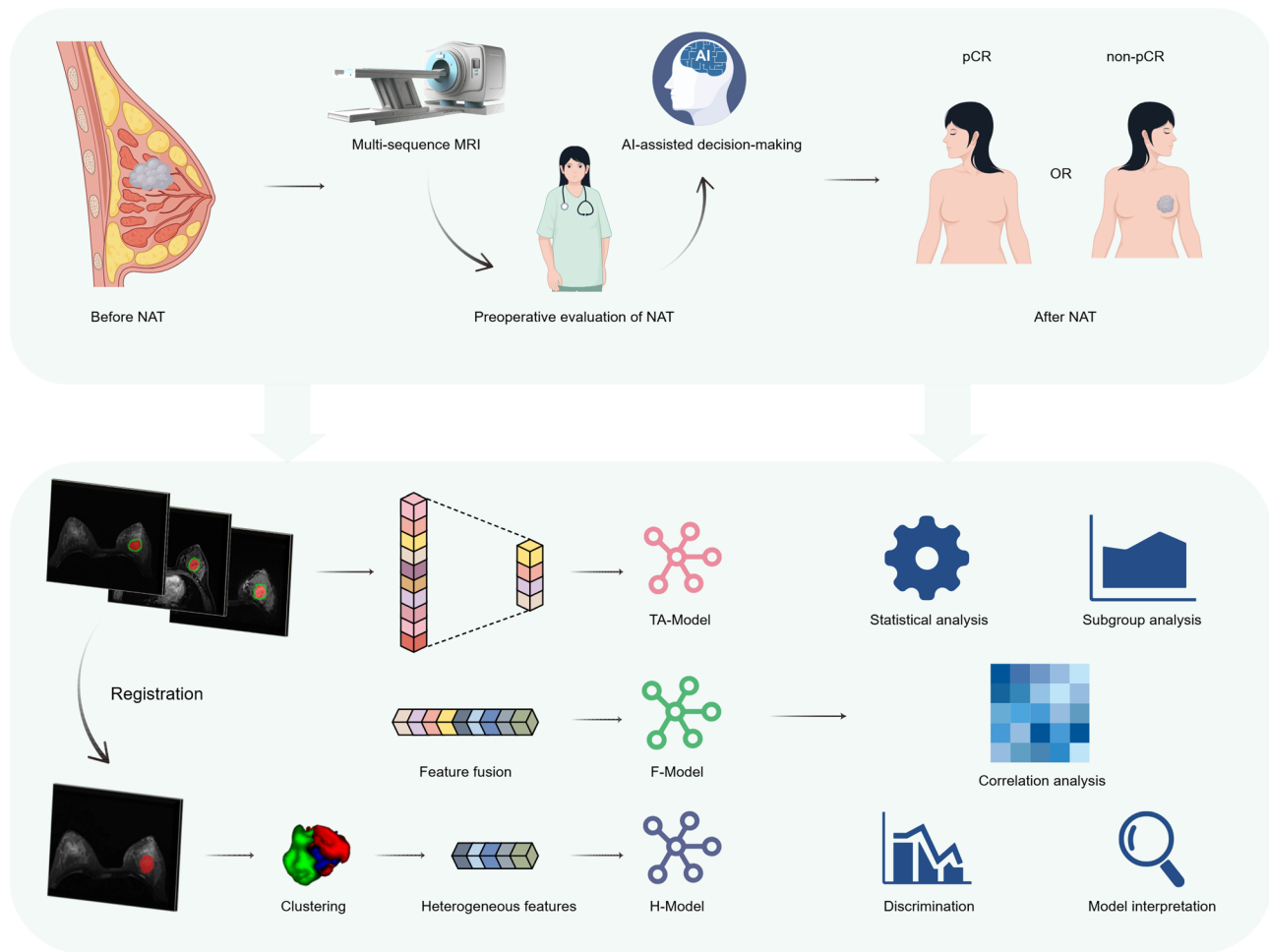


Fig. 1 The design and workflow of this study

the training set, while 41 were assigned to the test set. In addition, patients' clinical data was extracted and gathered from the electronic medical records of the hospital information system (HIS).

Breast MRI acquisition

All patients underwent full-protocol breast MRI examinations using a 3.0 Tesla MRI scanner (Skyra, Siemens Healthcare, Erlangen, Germany) equipped with a specialized 16-channel breast coil. Patients were scanned in a prone position, with both breasts naturally positioned within the coil. The coverage extended from the axilla to the lower margin of the breasts. The standard breast MRI acquisition technique includes fat-saturated T2-weighted imaging (T2WI), diffusion-weighted imaging (DWI), and T1-weighted fat-suppressed dynamic contrast-enhanced MRI (DCE-MRI). For DCE-MRI, a dual-barrel high-pressure injector was employed for intravenous administration of the contrast agent (Omniscan, GE Healthcare, Milwaukee, WI) with a dosage of 0.1 mmol/kg at a rate of 2.5 mL/s, followed by a 20 mL saline flush. Specific

imaging parameters are provided in Supplementary Table 1.

Pathological evaluation

The Miller–Payne (MP) grading system was utilized for the evaluation of pathological responses to NAT [21]. The Miller–Payne scoring encompasses five distinct grades: Score 1 indicates either no reduction or only minimal reduction in tumor size; Score 2 signifies a reduction of up to 30%; Score 3 represents a reduction spanning from 30 to 90%; Score 4 signifies a reduction exceeding 90%, accompanied by some remaining invasive (or axillary) disease; Score 5 designates the absence of any residual invasive carcinoma (or axillary) metastasis. MP Grades 1 through 4 are grouped under the category of pathological partial response (pPR), while MP Grade 5 is categorized as pCR.

Tumor segmentation

All T2WI, DWI, and DCE-DWI images were transported into the ITK-SNAP software (version 3.8.0. Available at

<http://www.itksnap.org>) to segment the tumor lesions manually. The whole lesion of each tumor was segmented by a radiologist with more than five years of experience in breast MRI interpretation while avoiding the regions of tumor necrosis, cysts, and calcifications. Subsequently, Python programming language (version 3.12.0. Available at <http://www.python.org>) was employed to expand the lesion ROIs outward by 3 mm, effectively embracing the peritumoral regions. Meanwhile, another radiologist with over ten years of experience in breast MRI interpretation randomly selected 50 patients for independent ROI delineation. These delineations were utilized to assess feature consistency and inter-observer variability analysis.

Feature engineering

To mitigate biases arising from variations in image field strengths and scanning protocols, preprocessing steps were applied to all images, including image normalization, N4 bias field correction, and resampling to an isotropic voxel size of $1 \times 1 \times 1$ mm³. T2WI, DWI, and DCE-MRI sequences were registered to create pre-clustered images. For each voxel within the tumor ROI, its intensity value in the pre-clustered image was used as the spatial coordinate, with equal weight assigned to each sequence. Clustering was performed using the K-Nearest Neighbors (KNN) algorithm ($k=3$), determined by the elbow method, resulting in three distinct tumor habitat regions.

Five metrics were extracted to characterize the clustered regions: Inertia, Calinski-Harabasz index (CH index), Silhouette coefficient, Separation, and Davies-Bouldin index (DB index), referred to as heterogeneity features. These metrics assess intra-cluster compactness, cluster quality, sample similarity, inter-cluster separation, and cluster similarity, respectively.

Radiomics features were extracted from the images using the open-source Pyradiomics module in Python, yielding 1197 features from T2WI, DWI, and DCE-MRI, including shape-based attributes, first-order statistics, texture descriptors, and transformation features. These features underwent mean-variance normalization, and interclass correlation coefficients (ICC) were calculated for reproducibility, excluding features with $ICC < 0.90$. For deep learning features, the largest cross-sectional ROI slice from each sequence was manually cropped and resampled to 400×400 pixels. The images were input into the ResNet50 model for training, extracting 2048 features per sequence from the final fully connected layer.

Dimensionality reduction techniques were applied according to feature type. Radiomics features were selected using the Mann-Whitney U test ($p < 0.05$) and correlation analysis (coefficients > 0.9). For deep learning features, Principal Component Analysis (PCA) with a threshold of 0.9 for explained variance was used. After

initial selection, the radiomics and deep learning features were combined into a composite feature matrix. Feature importance was ranked using six tree-based models, and the top 180 features from each model were intersected. Finally, recursive feature elimination (RFE) was applied, and the results were further refined through an additional intersection step.

Model construction and validation

The samples were divided into training and test sets using a random number table method. The model was built using the training set and evaluated using the test set. We constructed the tumor-adjacent model (TA-Model) using the optimal feature subset derived from the tumor and surrounding regions. Additionally, heterogeneity features were utilized to build the heterogeneity model (H-Model). We also developed a feature fusion model (F-Model) by combining the features from the individual models. For each type of model, we applied a variety of powerful classification algorithms, including Extreme Gradient Boosting (XGBoost), Random Forest (RF), Gradient Boosting Machine (GBM), Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost), Adaptive Boosting (AdaBoost), K-Nearest Neighbors (KNN), and Logistic Regression (LR).

The models were assessed based on the area under the receiver operating characteristic curve (AUC) along with its 95% confidence interval (CI). Furthermore, key performance metrics such as accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) were computed. To enhance the interpretability and clinical transparency of the model, we applied SHapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) after model development. SHAP, rooted in game theory, provides global feature importance by quantifying each feature's contribution to model predictions, allowing identification of key predictors. LIME, in contrast, builds local surrogate models to approximate the original model's behavior for individual cases, offering insight into patient-specific prediction logic. Together, these techniques facilitate understanding of the model's decision process and bridge the gap between algorithmic output and clinical interpretation. The workflow of this study is shown in Fig. 1.

Statistical analysis

The baseline data of the patients were analyzed using SPSS (version 26.0) and Python statistical packages. Continuous variables are presented as mean $\pm$ standard deviation, while categorical variables are expressed as frequencies and percentages. The Levene test was used to evaluate the homogeneity of variances. Inter-group differences were compared using the Mann-Whitney U test

or Student's t-test, as appropriate. The Chi-squared test or Fisher's exact test was applied to compare categorical variables. A two-sided p -value of less than 0.05 was considered statistically significant. The 95% confidence interval (CI) for the area under the curve (AUC) was calculated using the bootstrap method with 1,000 iterations. The DeLong test was performed to compare the AUCs of different models.

Results

Clinical baseline characteristics

A total of 203 patients were ultimately included in the study for further analysis (Fig. 1). The baseline clinical attributes of the patients included in this study are elaborated in Table 1. Out of the total patient population under investigation, 71 individuals were able to achieve a pCR following NAT. On the contrary, 132 patients still

Table 1 Baseline characteristics comparison between training and test sets

Variables	Test set($n=41$)	Training set($n=162$)	χ^2/t	PValue
Age (y)	45.93 $\pm$ 9.48	45.86 $\pm$ 10.82	0.037	0.159
BMI (kg/m ²)	23.37 $\pm$ 3.15	23.26 $\pm$ 2.68	0.237	0.283
CA153 (u/ml)	6.19 $\pm$ 14.98	22.64 $\pm$ 44.61	-0.911	0.156
cT stage, n (%)			0.805	0.669
cT1	4(21.05%)	15(78.95%)		
cT2	25(18.52%)	110(81.48%)		
cT3	12(24.49%)	37(75.51%)		
cN stage, n (%)			1.782	0.619
cN0	7(15.91%)	37(84.09%)		
cN1	17(22.08%)	60(77.92%)		
cN2	13(24.07%)	41(75.93%)		
cN3	4(14.29%)	24(85.71%)		
cM stage, n (%)			0.771	0.380
cM0	41(20.50%)	159(79.50%)		
cM1	0(0.00%)	3(100.00%)		
ER, n (%)			0.004	0.948
Negative	19 (20.00%)	76 (80.00%)		
Positive	22 (20.37%)	86 (79.63%)		
PR, n (%)			2.366	0.124
Negative	25(24.51%)	77(75.49%)		
Positive	16(15.84%)	85(84.16%)		
Her2, n (%)			0.487	0.485
Negative	26(21.85%)	93(78.15%)		
Positive	15(17.86%)	69(82.14%)		
Ki-67, n (%)			0.290	0.590
Negative	8(17.39%)	38(82.61%)		
Positive	33(21.02%)	124(78.98%)		
menopause, n (%)			0.016	0.899
Negative	26(20.47%)	101(79.53%)		
Positive	15(19.74%)	61(80.26%)		
pCR status, n (%)			0.095	0.758
Non-pCR	27(21.09%)	101(78.91%)		
pCR	15(19.74%)	61(80.26%)		

exhibited residual invasive BC or axillary lymph node metastasis post-treatment. The pCR rates for the training and test cohorts were 35.80% and 31.71% respectively. A comparative analysis of the clinical parameters between the training and test sets revealed no statistically significant differences for any clinical variables, including age, tumor size, and human epidermal growth factor receptor 2 (HER2) status. The detailed comparison of pCR distributions across molecular subtypes and neoadjuvant regimen groups is provided in Supplementary Table 2.

Feature selection

After the radiomics feature extraction, 1197 features were obtained from the T2WI, DWI, and DCE-MRI sequences. Following ICC consistency evaluation, 3214 features were retained (representing 89.5%). Mann-Whitney U test and correlation analysis were used for feature selection, resulting in 107 radiomics features. Deep learning features underwent PCA for dimensionality reduction, retaining 123 features. Finally, the dimension-reduced radiomics and deep learning features were integrated into a composite feature matrix, which was then used for feature selection through RFE. The intersection of the top 180 ranked features from each of the six tree-based models resulted in a total of 64 shared features (Fig. 2a). Figure 2b shows the application of RFE to fused features, aiming to select the optimal feature subset and reduce redundancy. The results indicate that most models stabilize in performance with around 10 features, with additional features providing limited improvement, highlighting 10 features as the key point for feature selection. However, these 10 features varied across different tree models, suggesting different feature preferences among the models. To address this, we conducted an intersection analysis of the selected features from the six tree models and identified four common features (Fig. 2c): DCE_LLL_DependenceUniformity, DWI_Log_Skewness, DWI_LHH_Skewness, and PC72. These features were defined as the final optimal subset, referred to as Tumor-Adjacent features.

Model construction and validation

Figure 3a compares the discriminative performance of various machine learning algorithms across the three models for predicting pCR. Figure 3b evaluates the performance of these algorithms based on multiple evaluation metrics. The results indicate that XGBoost consistently outperformed other algorithms in all three models. Consequently, XGBoost was selected for further model analysis and optimization.

Figure 4a presents a t-SNE visualization, providing an intuitive representation of the predicted distribution of samples by each model. The Decision Curve Analysis (DCA) in Fig. 4b evaluates the clinical utility

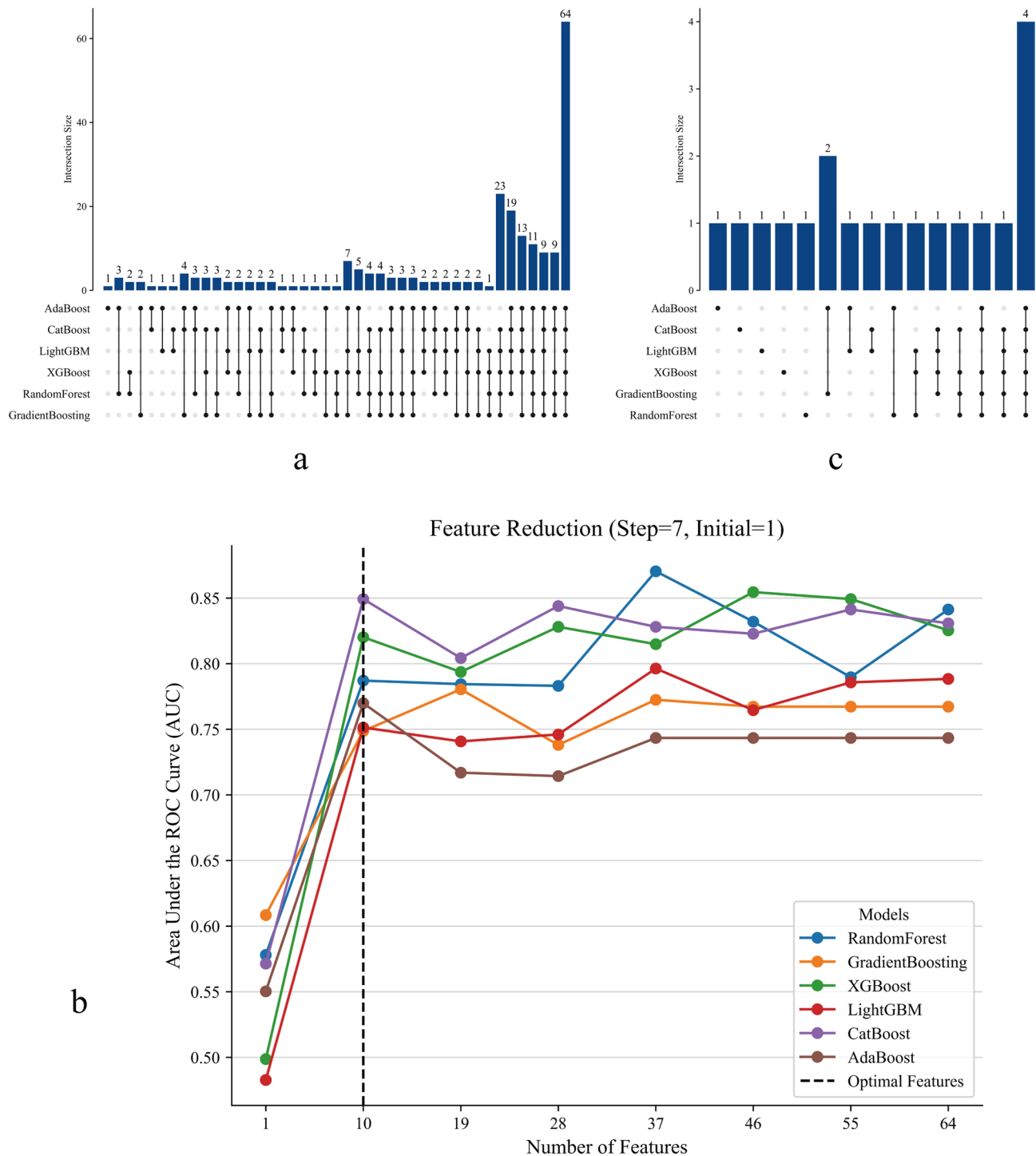


Fig. 2 (a) An upset plot showing the intersection of the top 180 ranked features in six tree-based models, (b) Recursive feature elimination of intersected features, (c) Intersection of the top 10 features after elimination

of the models, demonstrating that the F-Model yields the highest net benefit within clinically relevant threshold ranges. Figure 4c shows the Precision-Recall (PR) curves, highlighting the balance between precision and recall. Figure 4d displays the confusion matrices for the three

models, further illustrating differences in their classification performance.

Figure 5a illustrates the discriminative performance of the three models in predicting pCR, with the AUCs for the TA-Model, H-Model, and F-Model on the test set being 0.819 (95% CI: 0.625–0.963), 0.775 (95% CI:

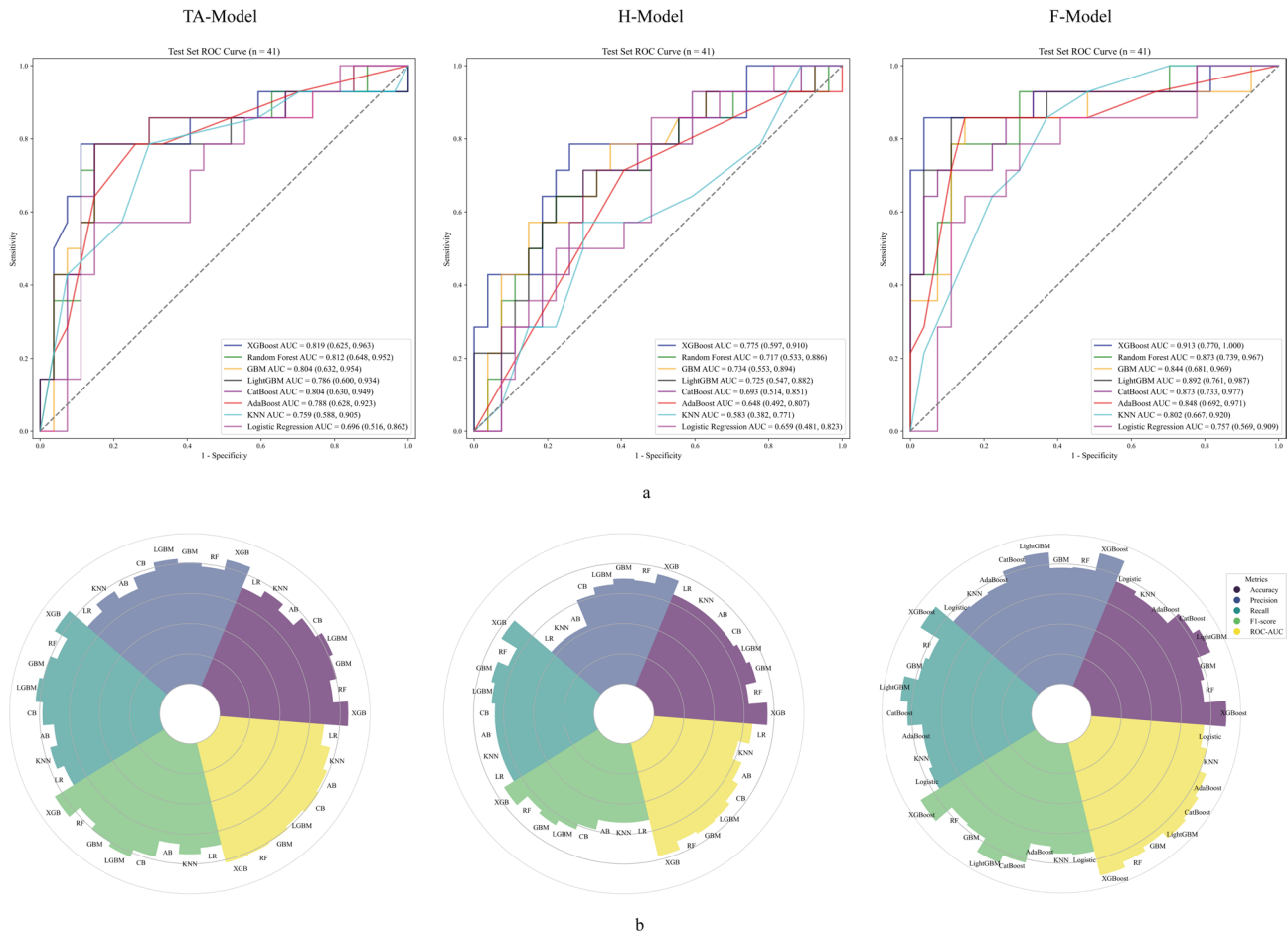


Fig. 3 (a) ROC curves for eight machine learning algorithms, (b) Radar chart comparing Accuracy, Precision, Recall, F1-score, and AUC metrics TA-Model, tumor-adjacent model; H-Model, heterogeneity model; F-Model, feature fusion model

0.597–0.910), and 0.913 (95% CI: 0.770–1.000), respectively. We also conducted a correlation analysis of the nine features used in the F-Model (Fig. 5b). A subgroup analysis was performed based on age (>45 vs. ≤45), molecular subtype (HER2+, HR+/HER2-, and TNBC), and pretreatment overall stage (stage II vs. stage III), as illustrated in Fig. 5c.

Machine learning models are often difficult to interpret, which is one of the major factors limiting their clinical application. The SHAP method provides global explanations at the feature level and local explanations at the individual level. Global explanations offer consistent and accurate attribution values for each feature, revealing the overall relationship between input features and target outcomes. As shown in the SHAP beeswarm plot (Fig. 6a), the average SHAP values are used to assess the contribution of each feature to the model, sorted in descending order. The results indicate that DWI_LHH_Skewness is the most critical feature for predicting pCR in BC patients. The heatmap (Fig. 6b) highlights the contribution of various features to pCR prediction, providing an overall view of the relationship between features and

individual samples. Figure 6c illustrates how the varying values of nine features influence the model's predictions, showing the direction and intensity of their impact.

Local explanations demonstrate how specific inputs affect the individual predictions of PTC patients. Waterfall plot displayed the contribution of each feature to the prediction for individual cases (X and Y), with positive and negative impacts shown in red and blue, respectively. Without feature information, the model's baseline value is -0.594. The LIME analysis (Fig. 7b and c) provides an alternative perspective; for Patients X and Y, the model predicts pCR probabilities of 26% and 71%, respectively, and demonstrates how different features influence the predictions for each patient. These local explanations enable clinicians to better understand and trust the model's predictions for individual patients, thereby enhancing the model's applicability in clinical settings.

Discussion

NAT combined with surgery has become an effective strategy for treating locally advanced BC. Early prediction of NAT efficacy is of significant clinical importance

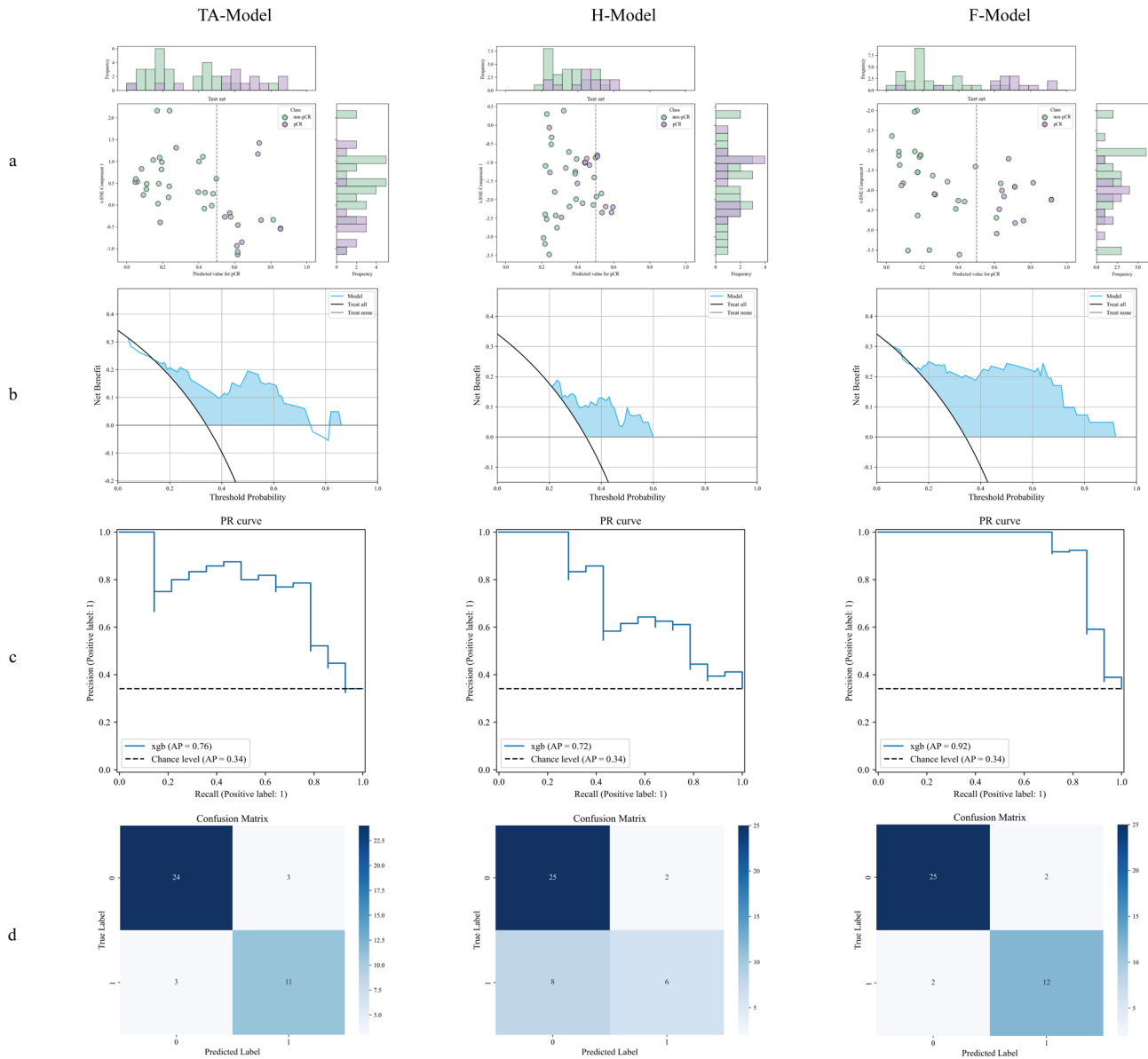


Fig. 4 Results of three XGBoost models, **(a)** t-SNE visualization, **(b)** DCA analysis, **(c)** PR curve comparison, **(d)** Confusion matrices for model predictions TA-Model, tumor-adjacent model; H-Model, heterogeneity model; F-Model, feature fusion model

for guiding treatment decisions and improving prognosis. However, no widely accepted non-invasive method for early prediction of pCR to NAT in BC exists. We developed an F-model based on multi-sequence MRI and multidimensional features (traditional radiomics and heterogeneous features) using machine learning algorithms. Our findings indicate that incorporating radiomics and heterogeneous features improves the model's predictive accuracy, providing a novel approach for the non-invasive early prediction of NAT efficacy in BC.

Histopathological examination of surgical specimens has traditionally been the gold standard for evaluating NAT efficacy in BC. However, it is limited by its retrospective nature, as it cannot provide real-time monitoring

or promptly reflect changes in tumor response. Currently, the Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 guidelines are commonly used, categorizing responses into four stages—complete response, partial response, stable disease, and progression—based on changes in tumor diameter [22]. However, this criterion is problematic as it overlooks the inherent growth characteristics and pre-treatment components of tumors. Consequently, relying solely on RECIST 1.1 criteria may not reflect the ability of anticancer therapy to induce some degree of tumor regression or growth retardation. Additionally, due to differences in tumor biology, the rate of tumor progression may vary not only between different types of malignancies but also among patients

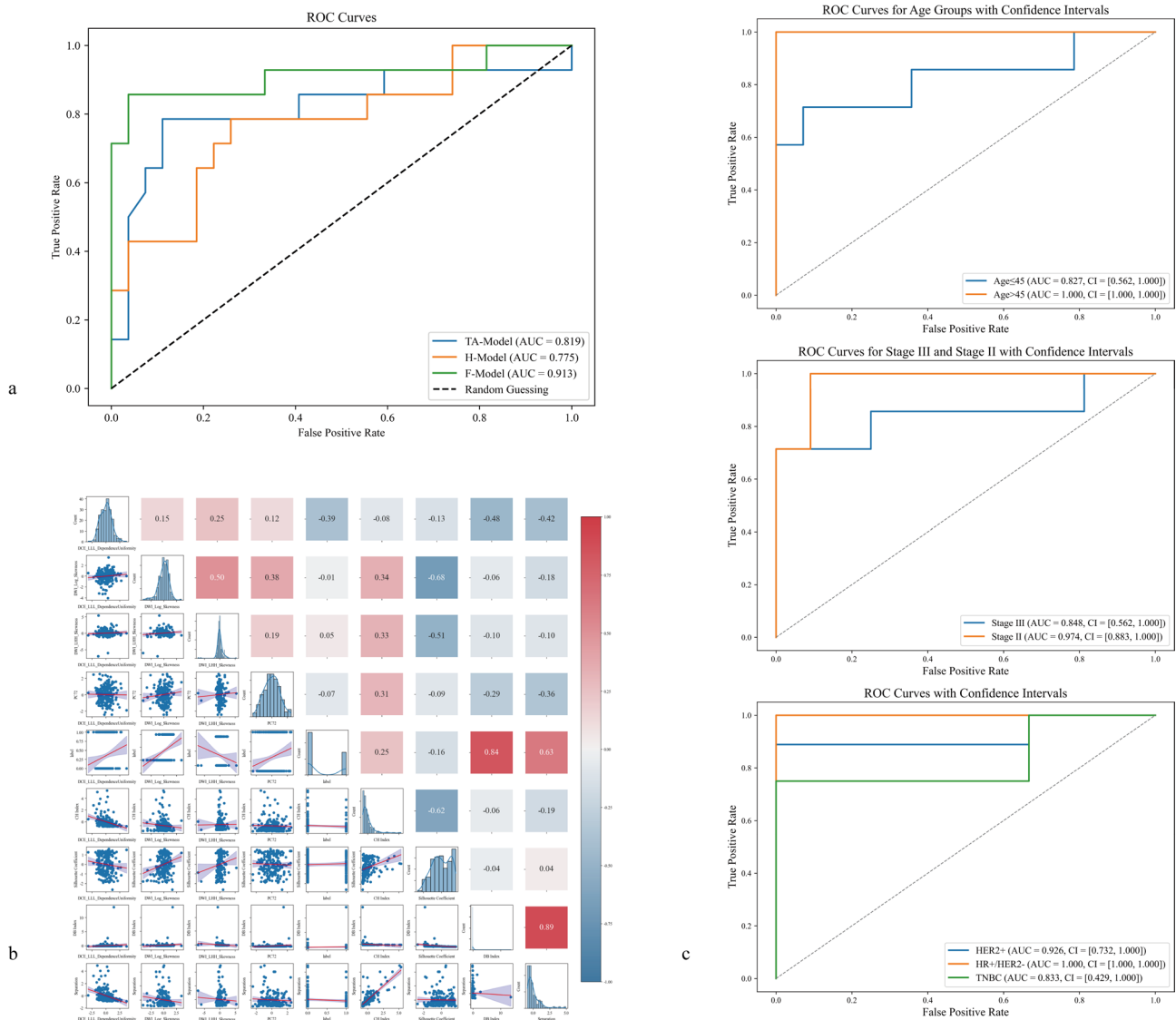


Fig. 5 (a) Comparison of ROC curves for three XGBoost models on the test set, (b) Correlation analysis of nine features in the F-model, (c) Subgroup analysis based on age, molecular subtype, and pre-treatment overall characteristics within the test dataset
TA-Model, tumor-adjacent model; H-Model, heterogeneity model; F-Model, feature fusion model

diagnosed with the same cancer [23, 24]. Empirical studies have further highlighted the limitations of relying solely on changes in the tumor's maximum diameter as a predictive metric of NAC efficacy. For example, research has indicated that the correlation between the residual tumor diameter shown on late-stage DCE-MRI in BC NAC and pathological findings is only 0.72, and the specificity of diagnosing pCR for non-enhanced lesions is only 0.54 [25, 26]. Moreover, some scholars have used MRI to assess the size of residual tumors in locally advanced BC patients after NAC and explored whether tumor morphology on MRI affects the accuracy of MRI measurements of residual tumor size. The results indicated an MRI-pathology correlation coefficient of 0.64, with 28% of patients showing inconsistencies between

MRI measurements and pathologically determined size. While MRI is an accurate method for measuring post-NAC residual tumor extent, it may overestimate residual lesions, especially in cases with a nest-like or margin tumor pattern, and in cases with ductal carcinoma in situ or multifocal dispersed nodules after NAC [27]. Collectively, these findings underscore the limitations of using maximum tumor diameter as the sole metric for NAC efficacy assessment. Changes in intratumoral heterogeneity are also key factors affecting efficacy, and traditional imaging examinations cannot provide quantitative assessment of tumor heterogeneity. Therefore, a more comprehensive approach, beyond changes in tumor diameter, is warranted for a more accurate and clinically relevant assessment of NAT efficacy in BC.

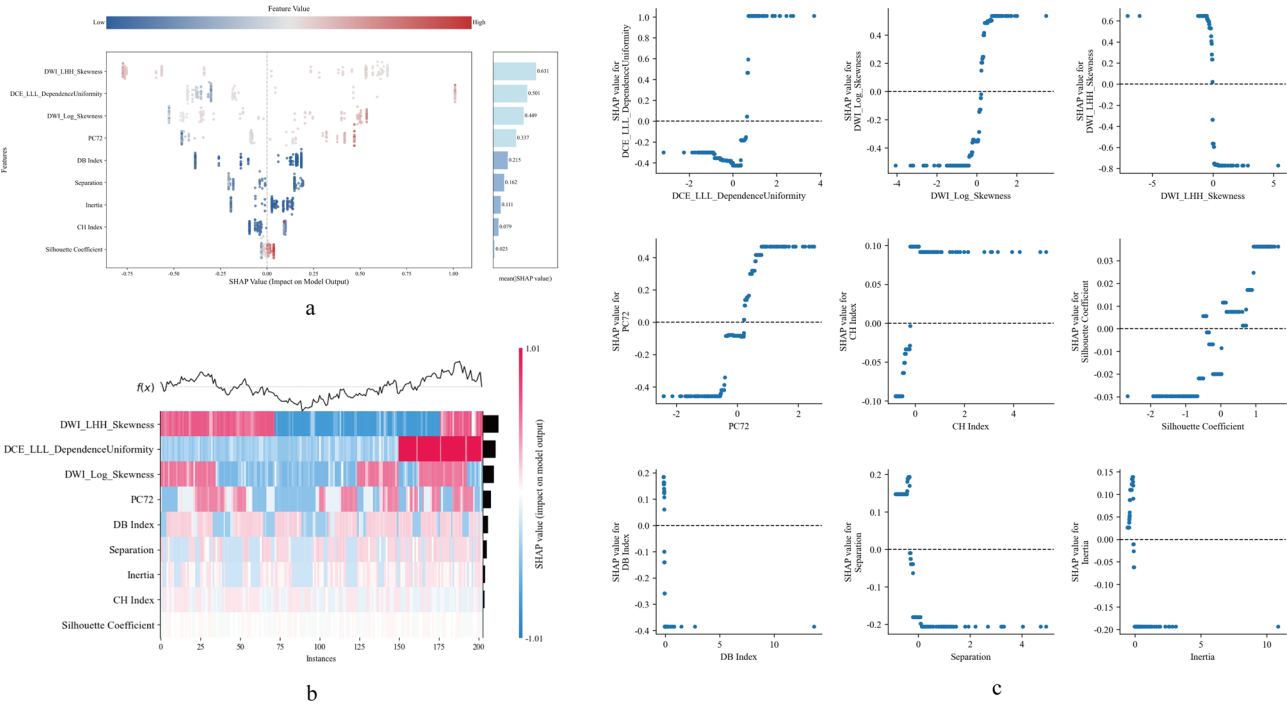


Fig. 6 SHAP analysis for interpreting F-Models in pCR prediction. (a) Honeycomb Plot (b) Heatmap (c) Scatter plot

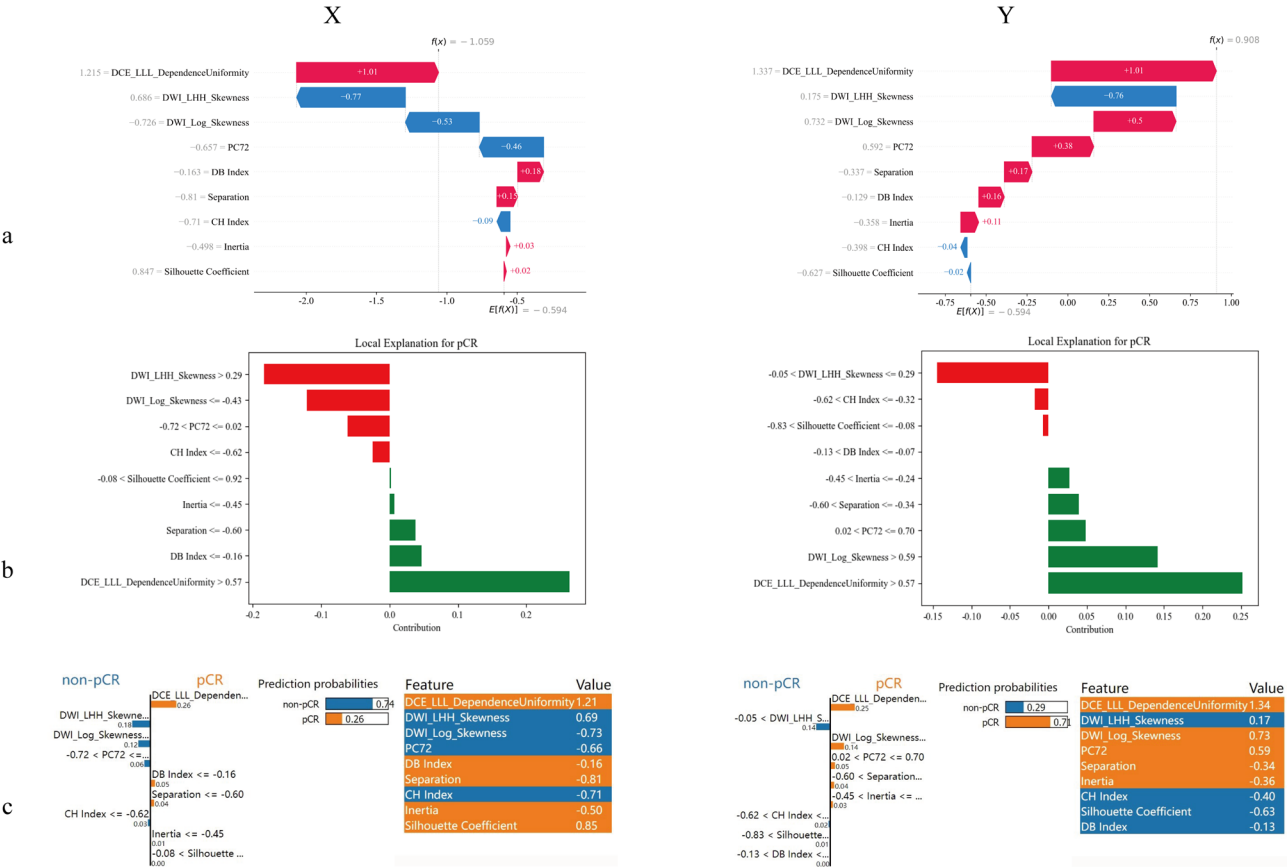


Fig. 7 SHAP and LIME analysis for pCR prediction in individuals X and Y using the F-model. (a) Waterfall plot (b) Local explanation bar chart (c) Comparison of prediction probabilities

BC's high heterogeneity and complexity lead to varying responses to NAT, with tumor heterogeneity evolving during treatment. Traditional imaging assessments based solely on maximum tumor diameter are insufficient to capture these dynamic changes. Recent studies suggest that internal homogeneity and perfusion index (PI%) from contrast-enhanced ultrasound may serve as early biomarkers for predicting pathological response to NAT in BC [28]. Sun et al. have demonstrated that Diffusion Kurtosis MR Imaging (DKI) could be a noninvasive approach for evaluating the effects of NAC in BC [29]. Other multidimensional approaches have been proposed, encompassing methods like 18 F FDG PET-CT, ultrasound elastography combined with serum ATF5, and the expression of LncRNA MIR205HG to refine NAC efficacy assessment [30–32]. To identify more effective biomarkers, some researchers have integrated artificial intelligence (AI) into assessing NAC efficacy in BC. Studies exploring radiomics and deep learning applications have shown promising results. For example, Li et al. combined FDG PET/CT radiomics with clinical features to predict pCR in NAC, achieving AUC values of 0.958 and 0.730 in the training and test sets, respectively, outperforming models based solely on clinical features [33]. Chen et al. analyzed MRI of 158 BC patients and developed radiomics features to predict pCR in NAC. Their model, combining pretreatment MRI features and hormone status, demonstrated high predictive value for NAC pCR [12]. ITH is a complex phenomenon arising from diverse cellular subpopulations within a tumor, each possessing distinct characteristics that ultimately lead to varying phenotypic and functional characteristics during disease progression [34]. This heterogeneity has been widely reported to contribute to treatment resistance and adverse clinical outcomes [35]. In the field of radiomics, conventional methods for quantifying intratumor heterogeneity include first-order feature histograms and second-order image features such as texture analysis [36]. These have shown value in assessing tumor heterogeneity [37, 38]. However, it should be noted that texture analysis methods presuppose homogeneity within the entire tumor and assume that the tumor is heterogeneous but well-mixed. This approach overlooks phenotypic variations in different regions within the tumor, potentially providing an incomplete quantification of intratumor heterogeneity [19]. To address this limitation, we utilized the “Habitat Imaging” technique, a more nuanced approach that identifies subregions—or “habitats”—within the tumor to better characterize its internal heterogeneity [19]. To represent tumor heterogeneity comprehensively, we delineated regions of interest in three dimensions, considering the entire tumor before subregion segmentation. This approach preserved spatial integrity and allowed for the exploration of relationships

between subregions in relation to the outcome variable. Using K-means clustering, we identified five heterogeneous features to build a machine learning model. These features effectively capture tumor subregion specifics, including intragroup similarities, and provide a detailed description of tumor heterogeneity. Our study highlights the importance of intratumor heterogeneity subregions in predicting treatment outcomes and prognosis, aligning with prior research [39–41].

Explainability analysis in radiomics helps to understand how models make predictions based on imaging features, enhancing transparency and trustworthiness. It reveals the relationship between imaging features and disease biology, while also aiding in model optimization and improving clinical decision support [42–44]. Overall, DWI_LHH_Skewness was the most critical feature for predicting pCR. This feature, derived from DWI imaging, reflects the asymmetry of pixel intensity distribution within the tumor region, highlighting tumor heterogeneity and structural differences. However, clinically, DCE-MRI is considered a reliable tool for monitoring residual tumors [45], which was also reflected in our individual analyses. Although the performance of the heterogeneity model alone was suboptimal, the fusion model demonstrated significantly improved efficacy. This indicates that heterogeneity features primarily serve a supportive role, enhancing the predictive performance when integrated with radiomics features.

While the findings of this study are promising, several limitations warrant attention. Firstly, the fusion radiomics model developed herein is predicated on retrospective data, which, although informative, may not wholly represent the complex landscape of BC heterogeneity. The external validation of this model using prospective data from randomized clinical trials would substantively augment its clinical validity.

Secondly, the study was confined to a single institution and encompassed a relatively limited sample size, thereby necessitating further validation through larger, multi-center studies to ensure its generalization. Thirdly, we exclusively utilized MRI data to develop the model. The incorporation of additional types of medical imaging, such as pathology whole-slide images or ultrasound scans, could further refine the model's predictive accuracy.

In conclusion, we leveraged multi-sequence and multidimensional radiomics features from pre-NAT MRI, and combined various machine learning algorithms to construct a fusion model for accurate prediction of pCR in BC. This model holds clinical value for BC patients, assisting surgeons in deciding whether to proceed with breast-conserving surgery and sentinel lymph node biopsy after NAT treatment. Additionally, we aim

to explore the clinical utility of tailored treatments in broader regions and populations.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40644-025-00929-2>.

Supplementary Material 1

Author contributions

Shaojie Xu was responsible for methodology, validation, and writing the original draft; Yushi Ying contributed to data curation and drafting the manuscript; Xingyin Li, Qilan Hu, Yulin Li, Hao Xiong, Yanyan Chen, and Qing Ye supported data curation; Xingrui Li provided resources and secured funding; Tao Ai handled review and editing; Yue Liu oversaw project administration and contributed to funding acquisition; Yaying Du conceptualized the study and led funding acquisition. All authors have read and approved the final version of the manuscript, confirm responsibility for all aspects of the work, and consent to its submission to this journal.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to data privacy and intellectual property issues but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study is a retrospective analysis and has received ethical approval from the Medical Ethics Committee of Huazhong University of Science and Technology Tongji Medical College Affiliated Tongji Hospital (Approval No: TJ-IRB20231173). The study was conducted in strict accordance with the ethical principles outlined in the Declaration of Helsinki. Given the retrospective nature of the study, the requirement for informed consent was waived by the ethics committee.

Competing interests

The authors declare no competing interests.

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